IoT devices, gateways, routers, etc. to connect	**Main considerations**	**Specific questions**	**Responses**
20 tank instrument sets; 4 × Arduino Opta RS485 controllers with A0602/D1608 I/O; 20 Reolink RLC-811A PoE cameras; TP-Link TL-SG3428XMP managed PoE+ switch; AAEON BOXER-8653AI edge computer; Teltonika RUTX11 router.	*Determine the connectivity between selected IoT devices, gateways, routers, etc.*	*Why do selected devices need to be connected with each other and to the network?*	Connectivity closes three loops: probes provide feedback to local control; controllers and cameras provide farm-wide context to the edge computer; selected events reach cloud users. It enables unit-specific alarms, verification of actuator outcomes, cross-unit comparison, nesting evidence and maintenance diagnostics. Safety interlocks remain in each Opta, so network loss reduces visibility rather than basic control.
	Determine communication and network technologies required to support connectivity	*What are the range and coverage requirements?*	Four RS-485 trunks each serve five nearby tanks and are deliberately kept below 100 m in a linear, terminated bus. Modbus guidance supports multidrop RS-485 and up to 32 unit loads per segment (Modbus Organization, 2006). Cat6 links from PLCs, cameras, edge computer and router remain within the 100 m copper-Ethernet channel limit. The control path does not depend on Wi-Fi coverage.
		What are the power consumption constraints?	Mains is available, so continuity and protection matter more than battery life. A protected 24 VDC supply powers probes, PLCs and valves; PoE+ powers cameras; separate AC circuits feed heaters and pumps. A UPS supports PLCs, switch, edge computer and router for alarm delivery, buffering and orderly shutdown. Contactors, RCDs and manual isolation protect high-current loads.
		What is the required data rate (how much data needs to be transmitted over time)?	Minute telemetry and state changes remain below 0.1 Mbit/s. Cameras dominate: 20 H.265 streams at about 4–8 Mbit/s each produce 80–160 Mbit/s aggregate. A gigabit backbone provides headroom. The edge analyses lower-rate substreams, keeps main streams local and uploads only selected event clips.
		What are the latency requirements (how fast does data need to be transmitted)?	PLC interlocks and heater/valve decisions target sub-second local response. Local alarms and dashboard state should update within about 5 s. Cloud reports tolerate seconds to minutes. No safety action waits for a cloud round trip; cloud commands are authenticated requests that the edge and PLC revalidate against local limits.
		What are the security and reliability requirements?	The switch separates control, camera and management VLANs and applies ACLs, 802.1X/MAC controls and QoS. Its 24 PoE+ ports and 384 W budget cover 20 cameras with reserve (TP-Link, n.d.). Devices use unique credentials; unused services and camera P2P/UPnP are disabled; administration uses VPN. MQTT uses TLS and X.509 certificates. UPS power, watchdogs, local buffering, normally closed valves and Ethernet-primary/LTE-failover WAN improve
		What are the cost considerations?	*Wired installation and industrial probes cost more initially than hobby sensors or Wi-Fi, but reduce battery labour, interference and replacement risk in a wet fixed site. One managed PoE switch removes 20 camera power adapters. Four zone PLCs limit I/O cable length. Edge filtering avoids continuous video upload and reduces WAN, cloud-ingest and storage costs.*
		Which communication/network technologies are selected?	Field level: 4–20 mA for analogue probes; RS-485/Modbus RTU for multidrop sensors; pulse/UART for simple instruments; isolated relay/24 V control for actuators. Plant level: 1000BASE-T Cat6 and IEEE 802.3af/at PoE through a TP-Link TL-SG3428XMP (TP-Link, n.d.). WAN: Ethernet primary plus Teltonika RUTX11 dual-SIM LTE Cat 6 failover (Teltonika Networks, n.d.). Applications use local Modbus TCP/HTTPS/RTSP and outbound MQTT/TLS/HTTPS.